Well	KN-104	**Rig**	Rig KD-3	**Report no**	20	**Date**	2011-03-21
Hole size	8½ in	**Depth 00:00**	3,364.3 m MD	**Depth 24:00**	3,530.4 m MD	**TVD 24:00**	3,530.2 m
Spud date	2011-03-02	**Days from spud**	20	**RT elevation**	123 m	**Field**	Kathal Nala

24 H SUMMARY

Drilled 8½ in hole from 3,364 m to 3,530 m, 166 m. NPT 5.4 h: partial losses at the barail base.

24 H FORECAST

Drill ahead 8½ in hole.

OPERATIONS

From	To	Hrs	Code	Depth m	Operation
00:00	03:23	3.38	DRL	3,391.5	Drilled 8½ in hole from 3,364 m to 3,392 m, avg ROP 8.1 m/h. WOB 17 t, 110 rpm, 920 gpm. Connections and MWD surveys each stand.
03:23	06:31	3.14	DRL	3,420.0	Drilled 8½ in hole from 3,392 m to 3,420 m, avg ROP 9.1 m/h. WOB 17 t, 110 rpm, 920 gpm. Connections and MWD surveys each stand.
06:31	10:23	3.86	DRL	3,448.5	Drilled 8½ in hole from 3,420 m to 3,449 m, avg ROP 7.4 m/h. WOB 17 t, 110 rpm, 920 gpm. Connections and MWD surveys each stand.
10:23	16:09	5.77	DRL	3,505.5	Drilled 8½ in hole from 3,449 m to 3,506 m, avg ROP 9.9 m/h. WOB 17 t, 110 rpm, 920 gpm. Connections and MWD surveys each stand.
16:09	18:34	2.41	DRL	3,530.4	Drilled 8½ in hole from 3,506 m to 3,530 m, avg ROP 10.3 m/h. WOB 17 t, 110 rpm, 920 gpm. Connections and MWD surveys each stand.
18:34	22:34	4.00	NPT	3,530.4	Partial losses of 28 bbl/h near the base of the coal-shale with ECD at 10.9 ppg.
22:34	24:00	1.44	NPT	3,530.4	Pumped a 40 bbl calcium carbonate pill at 25 ppb and cut flow to 780 gpm. Losses cured after 6 h.

INCIDENTS AND NPT

INCIDENT KN-104-E07 | LOSSES
Title: Partial losses at the Barail base
Start: 18:34 **Depth:** 3,530.4 m MD **MW:** 10.6 ppg **NPT:** 11.0 h
Description: Partial losses of 28 bbl/h near the base of the coal-shale with ECD at 10.9 ppg.
Action: Pumped a 40 bbl calcium carbonate pill at 25 ppb and cut flow to 780 gpm. Losses cured after 6 h.
Remarks: Losses began with ECD above 10.8 ppg. Hold ECD under 10.8 ppg approaching the Main Sand.

MUD PROPERTIES

Type	Mud weight	Funnel vis	PV	YP	Fluid loss
WBM, KCl-polymer-glycol	10.6 ppg	58 s	18 cP	17 lb/100 ft²	3.8 ml

BIT RECORD

Bit no	Size	Type	Depth in	Metres	Hours
8	8½ in	PDC, 7 blade	3,319.7 m	210.7	33.4

SURVEYS

MD m	Inc deg	Azi deg	TVD m
3,390.0	1.04	265.2	3,389.8
3,420.0	0.99	262.6	3,419.8
3,450.0	1.17	262.3	3,449.8
3,480.0	1.02	261.8	3,479.8
3,510.0	1.15	262.8	3,509.8